Use the following diagram to answer questions 8 and 9.

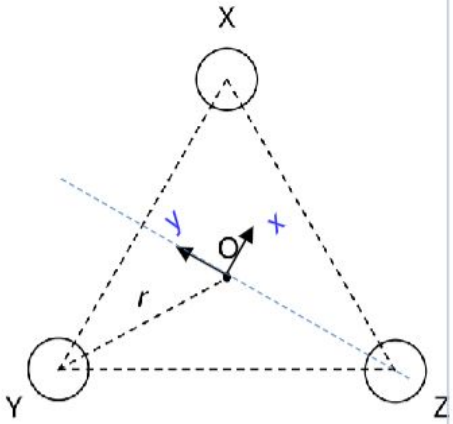
Three planets, X, Y and Z have masses *M*, *M* and *2M* respectively, are arranged in a triangle and are equidistant from each other.

Q8. Ans: D

Each planet contributes a gravitational field at O of magnitude:

$$g = \frac{GM}{r^2}, \quad g_Y = \frac{GM}{r^2}, \quad g_Z = \frac{G(2M)}{r^2} = \frac{2GM}{r^2}$$

Since the triangle is equilateral, the fields from X and Y point toward X and Y respectively. By symmetry, the **horizontal components** of X and Y's fields cancel each other out, and their **vertical components** (pointing upward, away from Z) add together.



The vertical component of each (X and Y) pointing away from Z:

$$g_X \cos 30° + g_Y \cos 30° = 2 \times \frac{GM}{r^2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}GM}{r^2}$$

The field from Z points **toward Z** (downward):

$$g_Z = \frac{2GM}{r^2}$$

Net field:

$$g_{net} = \frac{2GM}{r^2} - \frac{\sqrt{3}GM}{r^2} = \frac{GM(2-\sqrt{3})}{r^2} = \frac{G(2M-\sqrt{3}M)}{r^2}$$

Since $2 > \sqrt{3}$, the net field points **toward Z**.

8 What is the magnitude and direction of the gravitational field strength at point O?

	magnitude	direction
A	0	no direction
B	$\frac{GM}{r^2}$	towards X
C	$\frac{GM}{r^2}$	towards Z
D	$\frac{G(2M-\sqrt{3}M)}{r^2}$	towards Z

Q9. Ans: D